

numbers, but the “is a descendant of” relation is not a total order on the set of all people, since there are individuals neither of whom is descended from the other. A total relation that is transitive, but not necessarily either symmetric or antisymmetric, is a *total preorder*.

Exercises

B.2-1

Prove that the subset relation “ $\subseteq$ ” on all subsets of $\mathbb{Z}$ is a partial order but not a total order.

B.2-2

Show that for any positive integer n , the relation “equivalent modulo n ” is an equivalence relation on the integers. (We say that $a = b \pmod{n}$ if there exists an integer q such that $a - b = qn$.) Into what equivalence classes does this relation partition the integers?

B.2-3

Give examples of relations that are

- a.* reflexive and symmetric but not transitive,
- b.* reflexive and transitive but not symmetric,
- c.* symmetric and transitive but not reflexive.

B.2-4

Let S be a finite set, and let R be an equivalence relation on $S \times S$. Show that if in addition R is antisymmetric, then the equivalence classes of S with respect to R are singletons.

B.2-5

Professor Narcissus claims that if a relation R is symmetric and transitive, then it is also reflexive. He offers the following proof. By symmetry, $a R b$ implies $b R a$. Transitivity, therefore, implies $a R a$. Is the professor correct?

B.3 Functions

Given two sets A and B , a **function** f is a binary relation on A and B such that for all $a \in A$, there exists precisely one $b \in B$ such that $(a, b) \in f$. The set A is called the **domain** of f , and the set B is called the **codomain** of f . We sometimes write $f: A \rightarrow B$, and if $(a, b) \in f$, we write $b = f(a)$, since the choice of a uniquely determines b .

Intuitively, the function f assigns an element of B to each element of A . No element of A is assigned two different elements of B , but the same element of B can be assigned to two different elements of A . For example, the binary relation

$$f = \{(a, b) : a, b \in \mathbb{N} \text{ and } b = a \bmod 2\}$$

is a function $f: \mathbb{N} \rightarrow \{0, 1\}$, since for each natural number a , there is exactly one value b in $\{0, 1\}$ such that $b = a \bmod 2$. For this example, $0 = f(0)$, $1 = f(1)$, $0 = f(2)$, $1 = f(3)$, etc. In contrast, the binary relation

$$g = \{(a, b) : a, b \in \mathbb{N} \text{ and } a + b \text{ is even}\}$$

is not a function, since $(1, 3)$ and $(1, 5)$ are both in g , and thus for the choice $a = 1$, there is not precisely one b such that $(a, b) \in g$.

Given a function $f: A \rightarrow B$, if $b = f(a)$, we say that a is the **argument** of f and that b is the **value** of f at a . We can define a function by stating its value for every element of its domain. For example, we might define $f(n) = 2n$ for $n \in \mathbb{N}$, which means $f = \{(n, 2n) : n \in \mathbb{N}\}$. Two functions f and g are **equal** if they have the same domain and codomain and if $f(a) = g(a)$ for all a in the domain.

A **finite sequence** of length n is a function f whose domain is the set of n integers $\{0, 1, \dots, n-1\}$. We often denote a finite sequence by listing its values in angle brackets: $\langle f(0), f(1), \dots, f(n-1) \rangle$. An **infinite sequence** is a function whose domain is the set $\mathbb{N}$ of natural numbers. For example, the Fibonacci sequence, defined by recurrence (3.31), is the infinite sequence $\langle 0, 1, 1, 2, 3, 5, 8, 13, 21, \dots \rangle$.

When the domain of a function f is a Cartesian product, we often omit the extra parentheses surrounding the argument of f . For example, if we have a function $f: A_1 \times A_2 \times \dots \times A_n \rightarrow B$, we write $b = f(a_1, a_2, \dots, a_n)$ instead of writing $b = f((a_1, a_2, \dots, a_n))$. We also call each a_i an

argument to the function f , though technically f has just a single argument, which is the n -tuple $(a_1, a_2, \dots, a_n)$.

If $f: A \rightarrow B$ is a function and $b = f(a)$, then we sometimes say that b is the **image** of a under f . The image of a set $A' \subseteq A$ under f is defined by $f(A') = \{b \in B : b = f(a) \text{ for some } a \in A'\}$.

The **range** of f is the image of its domain, that is, $f(A)$. For example, the range of the function $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = 2n$ is $f(\mathbb{N}) = \{m : m = 2n \text{ for some } n \in \mathbb{N}\}$, in other words, the set of nonnegative even integers.

A function is a **surjection** if its range is its codomain. For example, the function $f(n) = \lfloor n/2 \rfloor$ is a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since every element in $\mathbb{N}$ appears as the value of f for some argument. In contrast, the function $f(n) = 2n$ is not a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since no argument to f can produce any odd natural number as a value. The function $f(n) = 2n$ is, however, a surjective function from the natural numbers to the even numbers. A surjection $f: A \rightarrow B$ is sometimes described as mapping A **onto** B . When we say that f is onto, we mean that it is surjective.

A function $f: A \rightarrow B$ is an **injection** if distinct arguments to f produce distinct values, that is, if $a \neq a'$ implies $f(a) \neq f(a')$. For example, the function $f(n) = 2n$ is an injective function from $\mathbb{N}$ to $\mathbb{N}$, since each even number b is the image under f of at most one element of the domain, namely $b/2$. The function $f(n) = \lfloor n/2 \rfloor$ is not injective, since the value 1 is produced by two arguments: $f(2) = 1$ and $f(3) = 1$. An injection is sometimes called a **one-to-one** function.

A function $f: A \rightarrow B$ is a **bijection** if it is injective and surjective. For example, the function $f(n) = (-1)^n \lfloor n/2 \rfloor$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}$:

$$0 \rightarrow 0,$$

$$1 \rightarrow -1,$$

$$2 \rightarrow 1,$$

$$3 \rightarrow -2,$$

$$4 \rightarrow 2,$$

⋮

The function is injective, since no element of $\mathbb{Z}$ is the image of more than one element of $\mathbb{N}$. It is surjective, since every element of $\mathbb{Z}$ appears as the image of some element of $\mathbb{N}$. Hence, the function is bijective. A bijection is sometimes called a *one-to-one correspondence*, since it pairs elements in the domain and codomain. A bijection from a set A to itself is sometimes called a *permutation*.

When a function f is bijective, we define its *inverse* f^{-1} as

$$f^{-1}(b) = a \text{ if and only if } f(a) = b.$$

For example, the inverse of the function $f(n) = (-1)^n \lfloor n/2 \rfloor$ is

$$f^{-1}(m) = \begin{cases} 2m & \text{if } m \geq 0, \\ -2m - 1 & \text{if } m < 0. \end{cases}$$

Exercises

B.3-1

Let A and B be finite sets, and let $f: A \rightarrow B$ be a function. Show the following:

- a. If f is injective, then $|A| \leq |B|$.
- b. If f is surjective, then $|A| \geq |B|$.

B.3-2

Is the function $f(x) = x + 1$ bijective when the domain and the codomain are the set $\mathbb{N}$? Is it bijective when the domain and the codomain are the set $\mathbb{Z}$?

B.3-3

Give a natural definition for the inverse of a binary relation such that if a relation is in fact a bijective function, its relational inverse is its functional inverse.

★ B.3-4

Give a bijection from $\mathbb{Z}$ to $\mathbb{Z} \times \mathbb{Z}$.

B.4 Graphs

This section presents two kinds of graphs: directed and undirected. Certain definitions in the literature differ from those given here, but for the most part, the differences are slight. Section 20.1 shows how to represent graphs in computer memory.

A **directed graph** (or **digraph**) G is a pair (V, E) , where V is a finite set and E is a binary relation on V . The set V is called the **vertex set** of G , and its elements are called **vertices** (singular: **vertex**). The set E is called the **edge set** of G , and its elements are called **edges**. Figure B.2(a) is a pictorial representation of a directed graph on the vertex set $\{1, 2, 3, 4, 5, 6\}$. Vertices are represented by circles in the figure, and edges are represented by arrows. **Self-loops**—edges from a vertex to itself—are possible.

In an **undirected graph** $G = (V, E)$, the edge set E consists of *unordered* pairs of vertices, rather than ordered pairs. That is, an edge is a set $\{u, v\}$, where $u, v \in V$ and $u \neq v$. By convention, we use the notation (u, v) for an edge, rather than the set notation $\{u, v\}$, and we consider (u, v) and (v, u) to be the same edge. In an undirected graph, self-loops are forbidden, so that every edge consists of two distinct vertices. Figure B.2(b) shows an undirected graph on the vertex set $\{1, 2, 3, 4, 5, 6\}$.

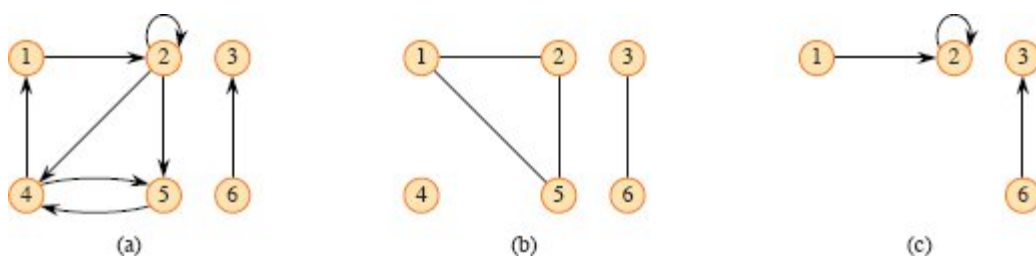


Figure B.2 Directed and undirected graphs. **(a)** A directed graph $G = (V, E)$, where $V = \{1, 2, 3, 4, 5, 6\}$ and $E = \{(1, 2), (2, 2), (2, 4), (2, 5), (4, 1), (4, 5), (5, 4), (6, 3)\}$. The edge $(2, 2)$ is a self-loop. **(b)** An undirected graph $G = (V, E)$, where $V = \{1, 2, 3, 4, 5, 6\}$ and $E = \{(1, 2), (1, 5), (2, 5), (3, 6)\}$. The vertex 4 is isolated. **(c)** The subgraph of the graph in part (a) induced by the vertex set $\{1, 2, 3, 6\}$.

Many definitions for directed and undirected graphs are the same, although certain terms have slightly different meanings in the two